underscores.

Note: We concatenated a String with an int: "The Sum is " + sum. The sequence is as follows: first the sum is converted from an integer to a string, which is then concatenated with the String "The Sum is: ". A common problem lot of people make is writing the code in the following way:

```
int number1 = 2;
int number2 = 4;
JOptionPane.showMessageDialog( null,
            "The Sum is: " + number1 + number2,
            "Screwy Result", JOptionPane.WARNING_MESSAGE
);
```

Error displayed by the code above

1.2 Data type

A variable called number1 actually refers to a place in memory where the value of the variable is stored. Every variable in Java has the following attributes: name, type, size, and a value.

1.2.1 Name

Variable names must begin with a letter. After that they can contain digits, the $ symbol and underscores. Java uses Unicode for its characters, so any "letter" that is valid for a word in any world language is valid for a name in Java.

1.2.2 Type

The "type" appears before the identifier name. It can be one of the "primitive data types" or it can be any previously defined class.

```
int num1;
```

This is a declaration. At this point, the name num1 refers to a location {a pointer} in the computer's RAM where this variable is stored. When an int is declared, four bytes are set aside. Still, nothing is stored in it yet.

1.2.3 Size

When we assign a type [int, String] to a variable, we are not only declaring a memory location. We also decide how big of a number or character is able to be stored in that variable.